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ML MODELLING SUPPORT

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ML Modelling Sample Report

Customer Churn Prediction

Branded demonstration sample for website download and client preview

Prepared for	Research Edit4u Solutions
Website	www.researchedit4u.in
Document type	Editable Word sample for PDF conversion
Note	All data in this sample are demonstration data created only for showcasing workflow and output quality.

This sample demonstrates how a raw customer dataset can be transformed into a deployable machine learning workflow with preprocessing, model selection, evaluation, interpretation, and business recommendations.

1. Executive Summary

Records	Target variable	Best model
3,500 customer records	Churn (Yes or No)	Logistic Regression (ROC-AUC 0.741)

This document is a full demonstration sample for the ML Modelling service offered by Research Edit4u Solutions. The sample uses a customer churn prediction use case because it is easy for non-technical clients to understand and also strong enough to demonstrate an end-to-end supervised machine learning workflow.

The modelling objective is to predict whether an existing customer is likely to leave the service. The workflow includes data understanding, missing-value treatment, feature preparation, exploratory pattern checks, model development, evaluation, interpretation of outputs, and practical deployment notes.

In this sample, four algorithms were compared: Logistic Regression, Decision Tree, Random Forest, and Gradient Boosting. The best-performing model on the hold-out test set was Logistic Regression, which achieved an ROC-AUC of 0.741, accuracy of 0.693, precision of 0.626, recall of 0.589, and F1-score of 0.607.

The sample also explains why each statistical or machine learning step was done, what business question it answers, and how the final results should be communicated to a client or stakeholder.

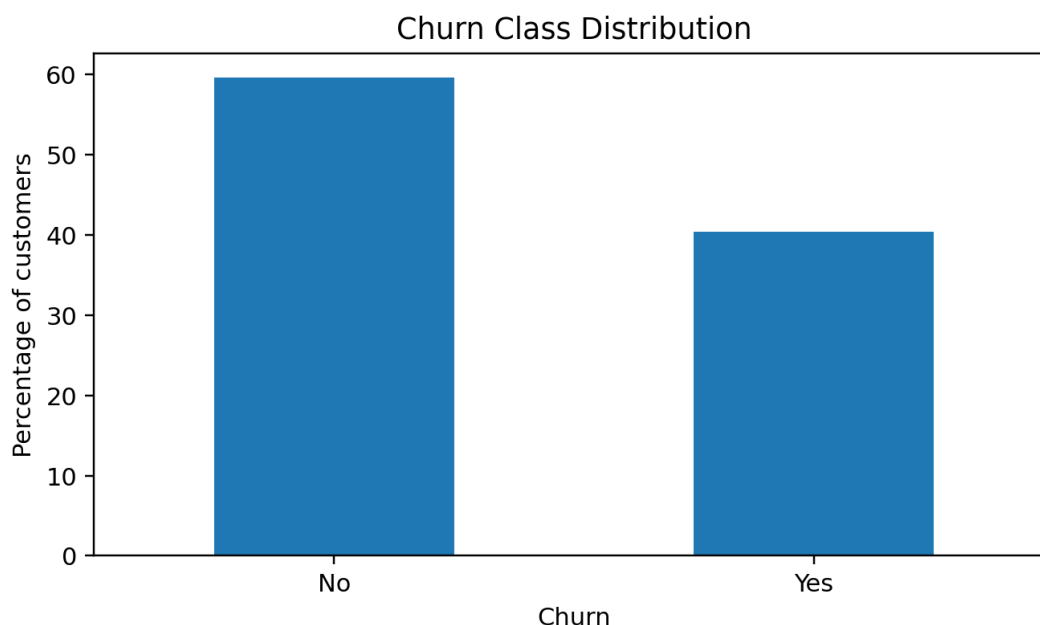


Figure 1. Distribution of the target variable in the sample dataset.

2. Business Problem, Objective, and Modelling Scope

Customer churn prediction helps an organisation identify customers who are at higher risk of leaving. Once those customers are identified, the business can intervene using retention offers, service recovery actions, support outreach, plan changes, or payment assistance.

Primary objective: To build a classification model that predicts whether a customer will churn using customer profile, account, service, billing, and interaction variables.

Secondary objectives: To identify the strongest churn drivers, compare multiple algorithms, and produce a practical output that can be used for decision making.

- Prediction task type: Binary classification
- Target variable: Churn, coded as Yes or No
- Expected deliverable to client: cleaned modelling dataset, model plan, evaluation table, interpretation notes, and prediction-ready output
- Business use case: prioritising retention actions for customers with the highest churn probability

3. Dataset Overview and Variable Dictionary

This sample dataset was created for demonstration purposes only. It simulates a telecom-style customer retention use case and contains structured fields that are commonly seen in churn projects. It is suitable for website showcasing because it is understandable, realistic, and safe to distribute.

Total records		3,500	
Total variables including target		14	
Predictor variables		12	
Numeric variables		5	
Categorical variables		7	
Observed churn rate		40.3%	
Variable	Type	Purpose in model	Expected relationship with churn
TenureMonths	Numeric	How long the customer has been with the service	Lower tenure usually increases churn risk
MonthlyCharges	Numeric	Current monthly bill size	Higher charges often increase churn risk
TotalCharges	Numeric	Cumulative value billed	Helps capture billing history and tenure effect

SupportTicketsLast6M	Numeric	Recent service pain points	More tickets may indicate dissatisfaction
LatePaymentsLast12M	Numeric	Payment stress or disengagement	Frequent late payments may increase churn
ContractType	Categorical	Commitment level of customer	Month-to-month customers often churn more
InternetService	Categorical	Service category used	Higher-complexity services may show more churn
PaymentMethod	Categorical	Billing behaviour pattern	Electronic check is often a churn risk signal
TechSupport	Categorical	Support plan coverage	No tech support may increase risk
PaperlessBilling	Categorical	Billing preference	Can correlate with digital-only segments
Partner	Categorical	Household context	Partnered customers may be more stable
SeniorCitizen	Categorical	Customer segment flag	May capture service sensitivity patterns

4. Data Quality Checks and Preprocessing Workflow

Preprocessing is one of the most important parts of a machine learning project. A strong sample should prove that the team does more than run algorithms. It should show that the raw data were inspected, cleaned, transformed, and prepared in a logically defensible way.

Column	Missing values (%)
MonthlyCharges	2.83
TotalCharges	3.00
PaymentMethod	2.83
TechSupport	2.77

- Missing value handling: Numerical fields were imputed using the median because the median is robust to skewed values and preserves central tendency without being too sensitive to extremes.

- Categorical missing values were imputed using the most frequent category. This is practical for service data where missingness is relatively low.
- Categorical encoding: One-hot encoding was used so that algorithms can work with non-numeric service categories without introducing artificial ordering.
- Feature scaling: Standard scaling was applied to numeric variables for algorithms that are sensitive to variable magnitude, especially Logistic Regression.
- Train-test split: The dataset was divided into 80% training data and 20% testing data using stratification so that the churn proportion remains similar in both subsets.

Subset	Rows
Training set	2800
Testing set	700

4.1 Why these preprocessing steps matter

Median imputation avoids losing records unnecessarily, which is important when client datasets are expensive to collect. One-hot encoding makes nominal categories usable in machine learning models. Standardisation gives fairer treatment to variables measured on very different scales. Stratified splitting gives a more trustworthy test set and reduces the chance of misleading results caused by an imbalanced split.

5. Exploratory Analysis and Screening Tests

Before model training, the sample includes a small set of screening analyses that help explain the data. These checks are useful in client reports because they show whether churn appears related to key variables even before advanced modelling is applied.

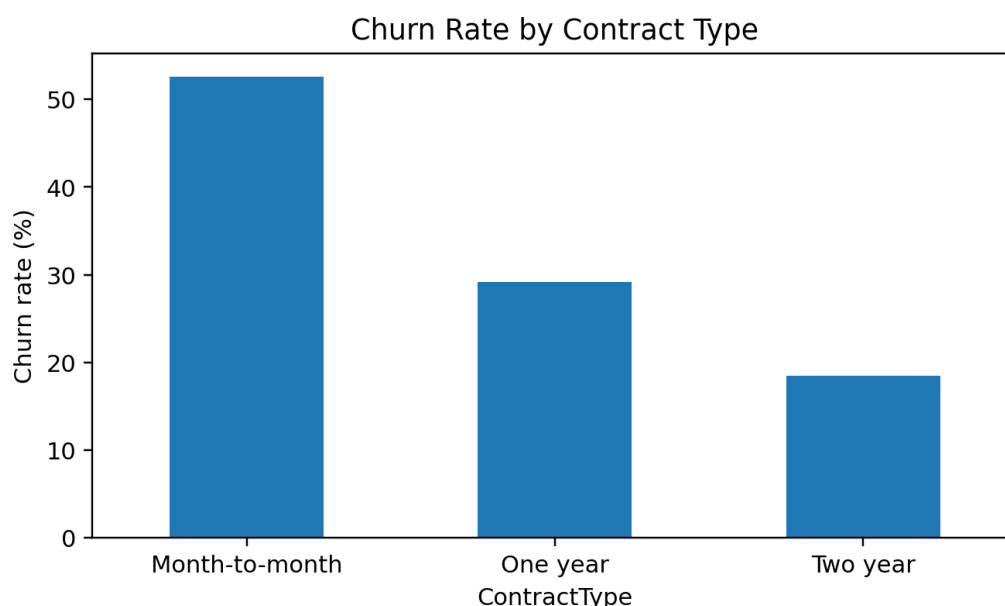


Figure 2. Churn rate differs strongly by contract type, with month-to-month contracts showing the highest risk.

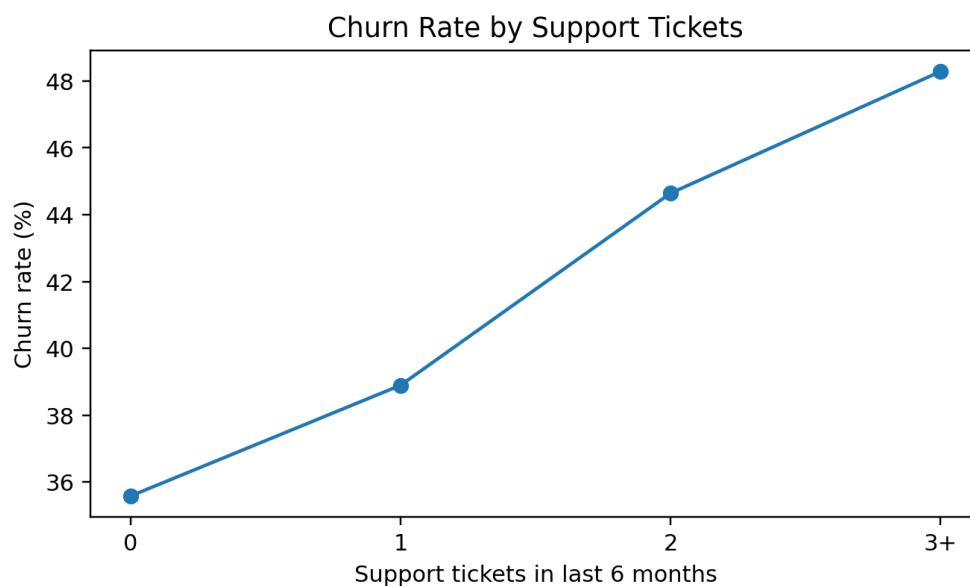


Figure 3. Churn rate increases as support ticket volume increases.

5.1 Chi-square tests for categorical predictors

The chi-square test checks whether a categorical predictor and the churn outcome are associated. It is useful because it tells us whether the distribution of churn differs meaningfully across categories such as contract type, payment method, or support status.

Variable	Chi-square	p-value
ContractType	297.56	0.0000
InternetService	34.57	0.0000
TechSupport	32.43	0.0000
PaperlessBilling	10.46	0.0012
PaymentMethod	17.03	0.0019
Partner	8.22	0.0042
SeniorCitizen	5.31	0.0212

Interpretation: Smaller p-values indicate stronger evidence that churn differs across categories. In this sample, variables such as ContractType, TechSupport, and InternetService showed clear association with churn and therefore were important candidates for the model.

5.2 Welch t-tests for numeric predictors

Welch t-tests compare the mean value of a numeric variable between churned and non-churned customers. Welch's version is used because it is safer when group variances may not be equal.

Variable	Mean if Yes	Mean if No	t-statistic	p-value
TenureMonths	31.37	39.24	-11.29	0.0000
MonthlyCharges	75.11	65.61	11.3	0.0000
SupportTicketsLast6M	1.35	1.14	5.28	0.0000
LatePaymentsLast12M	0.9	0.76	4.31	0.0000
TotalCharges	2339.53	2578.51	-3.97	0.0001

Interpretation: These tests show whether churned customers tend to have systematically different numeric profiles. In this sample, churned customers had lower tenure, higher monthly charges, and more service tickets on average.

5.3 Multicollinearity check using VIF

Variance Inflation Factor, or VIF, helps assess whether numeric predictors are too highly correlated with one another. Very high VIF values can make coefficient-based models unstable and difficult to interpret.

Variable	VIF
TotalCharges	7.92
TenureMonths	6.31
MonthlyCharges	2.82
SupportTicketsLast6M	1.0
LatePaymentsLast12M	1.0

Interpretation: VIF values close to 1 indicate low multicollinearity. In this sample, the values remain at acceptable levels for a demonstration model. If a live client dataset showed much larger VIF values, the workflow would include feature reduction or variable redesign.

6. Model Development Strategy

A robust ML sample should compare multiple algorithms instead of showing only one. This communicates methodological maturity and gives the client confidence that the selected model was chosen based on evidence rather than convenience.

Algorithm	Why included in the sample	When it is useful
Logistic Regression	Strong baseline model with interpretable coefficients	Best when interpretability matters and relationships are mostly linear on the log-odds scale
Decision Tree	Easy to explain with if-then logic	Useful when stakeholders want simple decision rules
Random Forest	Ensemble model that reduces overfitting from a single tree	Useful when nonlinear effects and interactions are expected
Gradient Boosting	Sequential ensemble that often captures subtle patterns	Useful when predictive performance is prioritised

- Training objective: maximise discrimination between churn and non-churn customers
- Primary comparison metric: ROC-AUC because it evaluates ranking ability across thresholds
- Secondary metrics: accuracy, precision, recall, and F1-score
- Selection logic: choose the model with the strongest overall balance of discrimination and operational usefulness

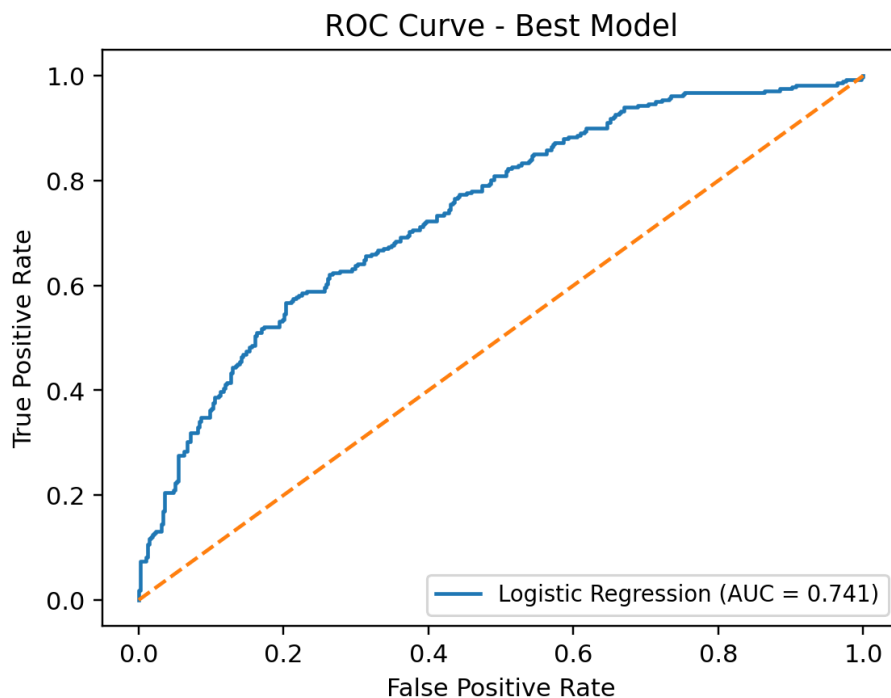
7. Model Evaluation Results

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	0.693	0.626	0.589	0.607	0.741
Random Forest	0.68	0.626	0.511	0.562	0.725
Gradient Boosting	0.68	0.612	0.56	0.585	0.725
Decision Tree	0.674	0.599	0.578	0.588	0.71

Model selection result: Logistic Regression provided the best ROC-AUC in the hold-out test sample and was therefore selected as the final demonstration model.

7.1 Meaning of each evaluation metric

- Accuracy: the overall proportion of correct predictions. It is intuitive but can be misleading if classes are imbalanced.
- Precision: among customers predicted to churn, the proportion that truly churned. This matters when retention actions are costly.
- Recall: among customers who truly churned, the proportion correctly identified by the model. This matters when missing at-risk customers is expensive.
- F1-score: the harmonic mean of precision and recall. It is useful when both false positives and false negatives matter.
- ROC-AUC: the probability that the model ranks a randomly chosen churned customer above a randomly chosen non-churned customer. This is a strong threshold-independent summary measure.



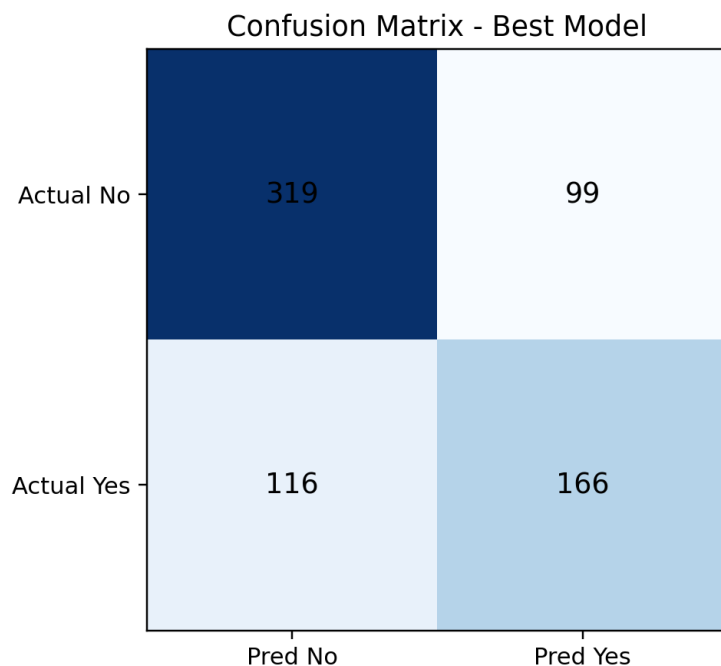


Figure 4. ROC curve and confusion matrix of the selected model.

8. Interpretation of the Final Model

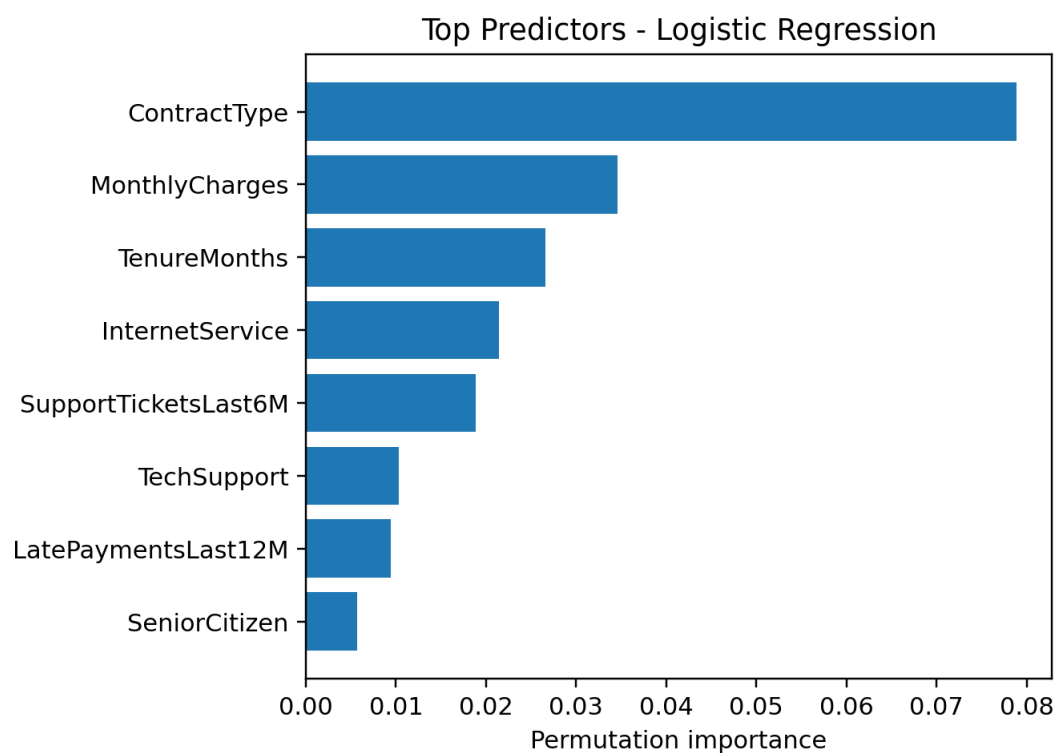


Figure 5. Most influential predictors in the final churn model.

The feature importance analysis suggests that the most influential variables in this sample are ContractType, MonthlyCharges, TenureMonths, InternetService, SupportTicketsLast6M. This aligns well with business intuition because churn is often driven by a combination of commitment level, billing burden, service friction, and account behaviour.

Business interpretation example: A newer month-to-month customer with high monthly charges, repeated support tickets, and no technical support plan is more likely to churn than a long-tenure customer on a longer contract with fewer support issues.

8.1 How the confusion matrix should be explained to a client

True negatives are stable customers correctly classified as non-churn. True positives are churn-risk customers correctly detected by the model. False positives are customers flagged as churn-risk who may not actually leave, which can increase retention campaign cost. False negatives are the most critical misses because these are customers who leave even though the model did not flag them.

9. What the Client Would Receive

Deliverable	Purpose	Typical format
Cleaned modelling dataset	Ensures transparency and reproducibility of inputs	Excel or CSV
Data dictionary	Explains every variable and coding choice	Word or PDF
Model comparison table	Shows how algorithms were evaluated	Word, Excel, or slide
Final model interpretation note	Translates outputs into simple business meaning	Word or PDF
Prediction output sheet	Lists each customer with churn probability and risk flag	Excel
Method summary and assumptions	Documents preprocessing, split strategy, metrics, and limitations	Word or PDF

This structure makes the service commercially clear. A client is not only paying for an algorithm. The client is paying for a documented workflow, an interpretable result, and a decision-ready output.

10. Sample Prediction Output Preview

TenureMonths	MonthlyCharges	SupportTicketsLast6M	ContractType	TechSupport	ActualChurn	PredictedChurn	ChurnProbability
54	63.84	3	Month-to-month	No	Yes	Yes	0.55
66	69.39	2	Month-to-month	Yes	No	No	0.274
3	54.84	1	Month-to-month	No	Yes	Yes	0.776
66	36.58	0	Month-to-month	Yes	Yes	No	0.115
50	69.84	2	Month-to-month	No	No	Yes	0.565
30	96.5	2	One year	Yes	No	No	0.41
32	112.13	2	Month-to-month	No	No	Yes	0.701
19	20.0	2	One year	Yes	No	No	0.109
25	103.77	2	Month-to-month	No	Yes	Yes	0.834
37	45.44	0	One year	Yes	No	No	0.093

In a live project, the final output sheet would normally include customer identifiers, probability scores, threshold labels such as low, medium, and high risk, and a recommendation column for action owners.

11. Limitations, Assumptions, and Future Scope

- This sample uses demonstration data. A real client dataset may need additional handling for outliers, leakage, imbalanced classes, or date-based validation.
- Threshold selection should be aligned with operational cost. A business with a high cost of losing customers may prefer a threshold that improves recall, even if precision drops.
- A production project may also include hyperparameter tuning, calibration, cross-validation, SHAP-based interpretation, and deployment through dashboards or APIs.
- If the client has monthly data over time, time-based validation is preferred over a simple random split.

Appendix A. Method Summary in Plain English

Question	Plain-English answer
Why did we split the data into train and test sets?	To make sure the model is evaluated on unseen data rather than the same data used for learning.
Why compare several models?	Different algorithms learn patterns differently. Comparing them helps identify the most suitable approach.
Why not rely only on accuracy?	A model can appear accurate while still missing many churn cases. That is why precision, recall, F1-score, and ROC-AUC were also used.
Why explain feature importance?	Clients need to know not only which model won but also which factors seem to drive risk.

THANK YOU

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